

Feature Proposal Document for Team November

(1) Feature Request: Forfeit Option for Tic-Tac-Toe and Connect 4

- **Description:** Implement a forfeit option that allows players to surrender a game of Tic-Tac-Toe or Connect 4 if they no longer wish to continue. This option should be easily accessible during gameplay and ensure that the opponent is notified of the forfeit result.
- **Expected Impact:**
 - This feature will provide players with a way to exit games gracefully rather than leaving them unfinished.
 - It improves the user experience by preventing unnecessary delays when a player wants to concede.
 - Additionally, it helps in ranking-based systems by properly registering the outcome of a forfeited game.
- **Implementation/Suggestions:**
 - Add a "Forfeit" button in the game interface, allowing players to voluntarily surrender the match.
 - Upon forfeiting, display a confirmation prompt to prevent accidental forfeits.
 - When a player forfeits, the opponent is automatically declared the winner, and the match result is recorded accordingly.
 - Notify both players when a forfeit occurs, displaying a message such as "Player X has forfeited the game. Player Y wins!"
 - In rankings, ensure forfeits are properly accounted for in win/loss records.

(2) Feature Request: Multi-Factor Authentication (MFA) for Enhanced Security

- **Description:** Introduce an optional multi-factor authentication (MFA) feature to improve account security. This would require users to verify their identity using a second authentication method, such as a one-time password (OTP) sent via email or mobile device, in addition to their password.
- **Expected Impact:**
 - Protects user accounts from unauthorized access, even if credentials are leaked.
 - Reduces fraud and increases trust in the platform's security.

- Aligns with modern cybersecurity best practices for account protection.
- **Implementation/Suggestions:**
 - Users can enable or disable MFA from their profile settings.
 - If enabled, after entering their password, users must enter an OTP sent via email or SMS.
 - Add a “Remember this device” option for users who don’t want to authenticate every time.
 - Implement a backup method in case users lose access to their email/phone.